

Champion circuits: symbolic Quantikz view, Stages A-G

BLUE boxes = encoding gates ($\text{angle} = a * x_i + b$, data-dependent)

ORANGE boxes = ansatz / learning gates (trained scalar)

control/target idioms (cx, cz) keep their native look

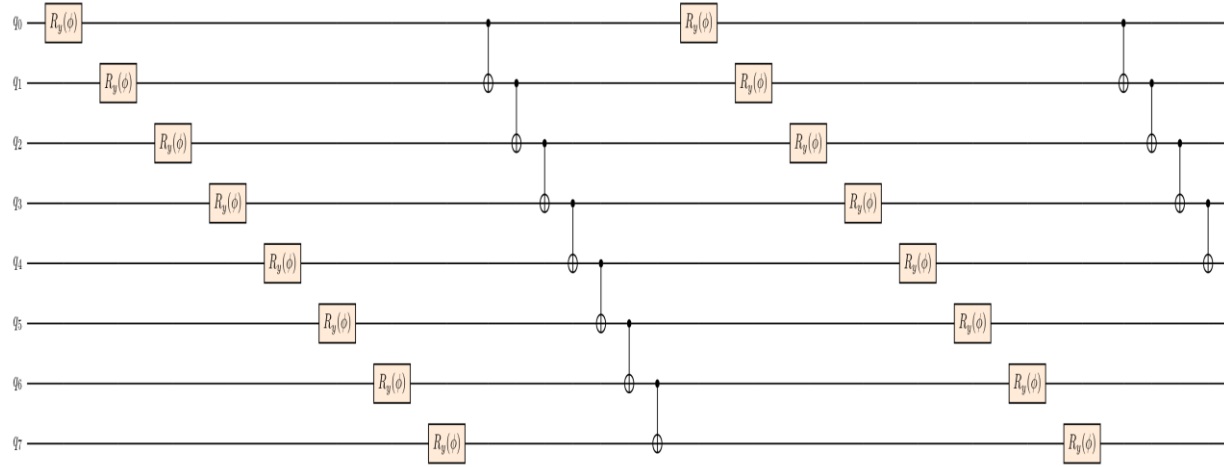
Look for interleaved blue/orange in Stages E/F (data re-uploading)
and clean block separation in Stage G (block-enforced).

How encoding and learning gates relate per stage

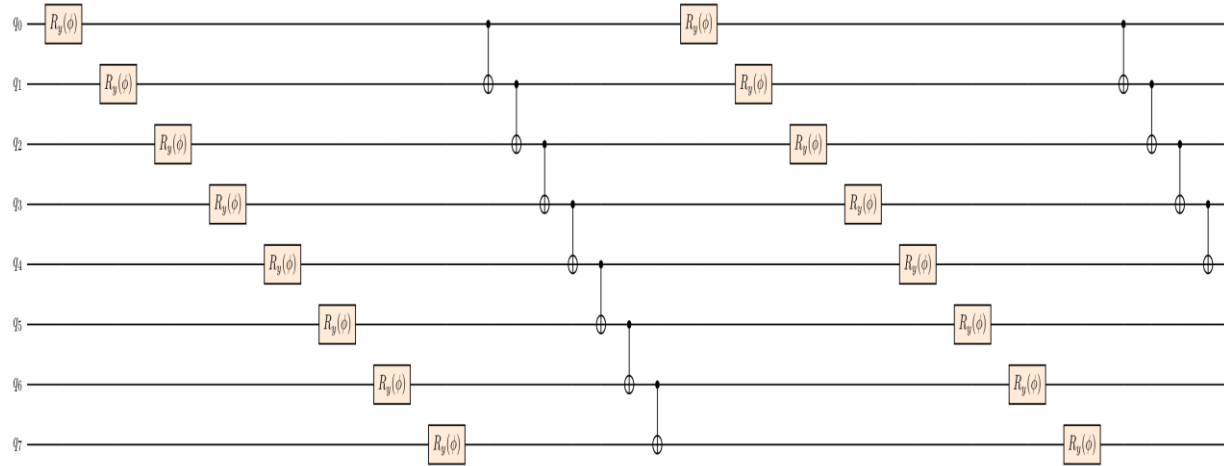
- A encoder runs in Python (external block) ; ansatz = fixed RY-CNOT chain
- B encoder in Python (block) ; ansatz = evolved genome
- C encoder in Python (block) ; ansatz = evolved genome ; evolved N
- D encoder classes (linear_proj, reupload_euler), consumed by B/C
- E v1/v2 encoder + ansatz gates inside ONE genome, sorted by depth -- INTERLEAVED
- F v1 enc_rot universal-1q (blue) + scalar Ising entanglers (orange) -- INTERLEAVED
- F v2 enc_rot + enc_xx/yy/zz feat-dep (blue) + ansatz (orange) -- INTERLEAVED (data reuploading)
- G same vocab as F v2, but qnode ENFORCES encoder-pass then ansatz-pass

color coding inside the per-stage slides: BLUE = encoding gates (data-dependent) ORANGE = ansatz / learning gates (trained scalar)

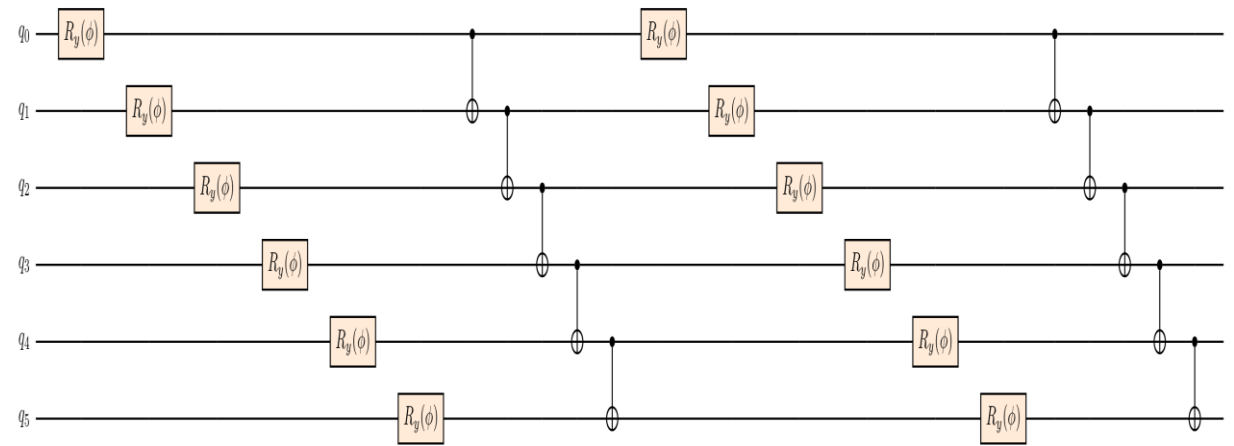
Stage A: Quantikz symbolic (ansatz only; external encoder noted in source .tex)



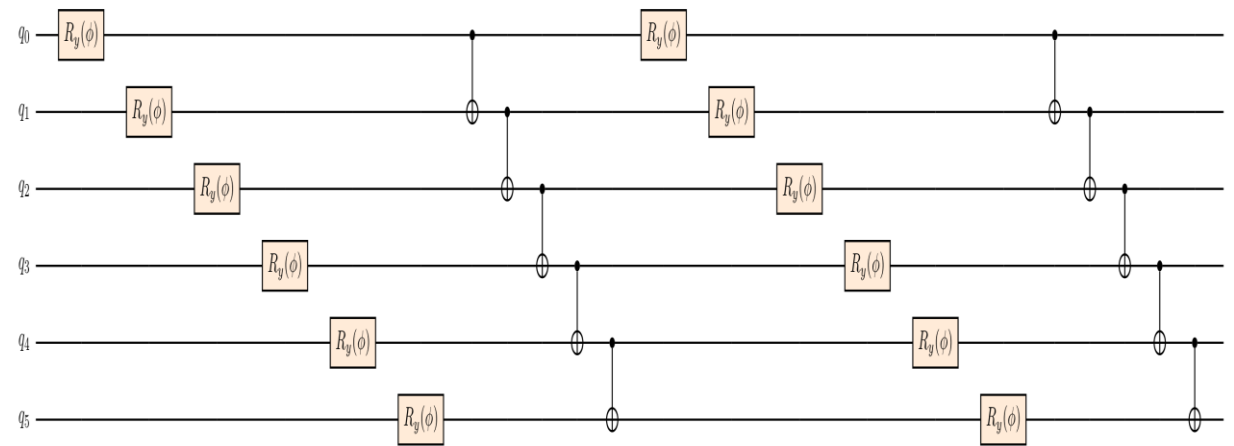
iris N=8 test_acc=0.967 encoder=learned



seeds N=8 test_acc=0.881 encoder=learned

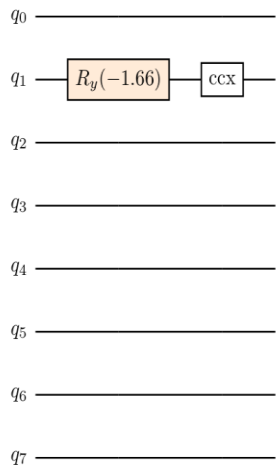


wine N=6 test_acc=0.833 encoder=learned



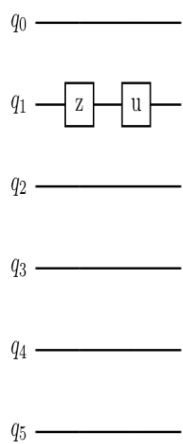
breast_cancer N=6 test_acc=0.895 encoder=learned

Stage B: Quantikz symbolic (ansatz only; external encoder noted in source .tex)



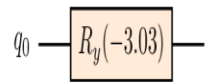
 Encoding gates (data-dependent: $a x_i + b$)  Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

iris N=8 test_acc=1.000 encoder=reupload_euler



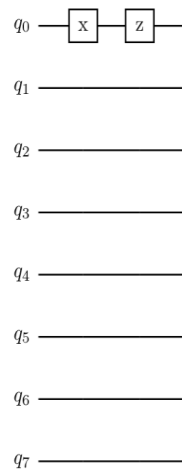
 Encoding gates (data-dependent: $a x_i + b$)  Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

seeds N=6 test_acc=1.000 encoder=reupload_euler



 Encoding gates (data-dependent: $a x_i + b$)  Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

wine N=4 test_acc=1.000 encoder=reupload_euler



 Encoding gates (data-dependent: $a x_i + b$)  Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

breast_cancer N=8 test_acc=0.974 encoder=reupload_euler

Stage B 3-seed champion (Quantikz)



iris N=10 test_acc=1.000 encoder=reupload_euler



seeds N=10 test_acc=1.000 encoder=linear_proj

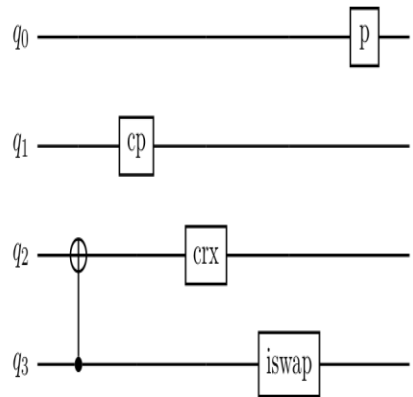


wine N=8 test_acc=1.000 encoder=reupload_euler



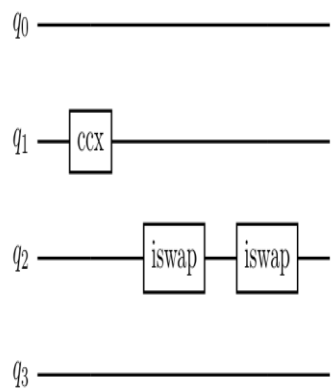
breast_cancer N=6 test_acc=0.974 encoder=reupload_euler

Stage C: Quantikz symbolic (ansatz only; external encoder noted in source .tex)



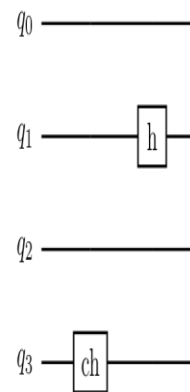
Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

iris N=4 test_acc=1.000 encoder=reupload_euler



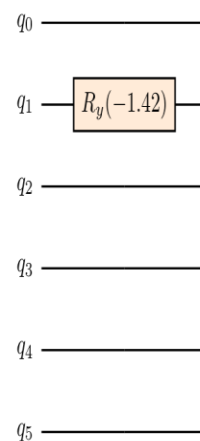
Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

seeds N=4 test_acc=1.000 encoder=reupload_euler



Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

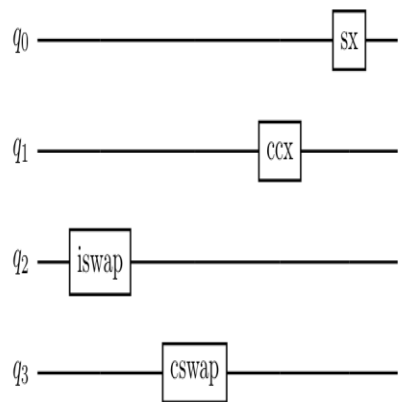
wine N=4 test_acc=1.000 encoder=reupload_euler



Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

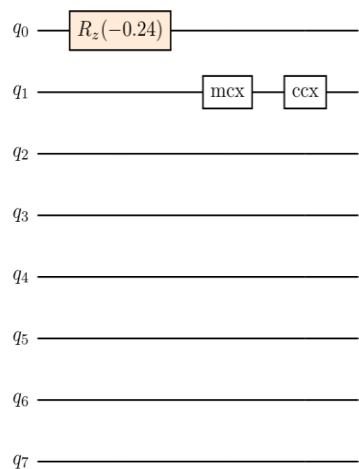
breast_cancer N=6 test_acc=0.974 encoder=reupload_euler

Stage C 3-seed champion (Quantikz)



Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

iris N=4 test_acc=1.000 encoder=reupload_euler



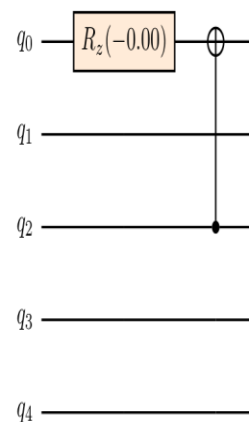
Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

seeds N=8 test_acc=1.000 encoder=reupload_euler



Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

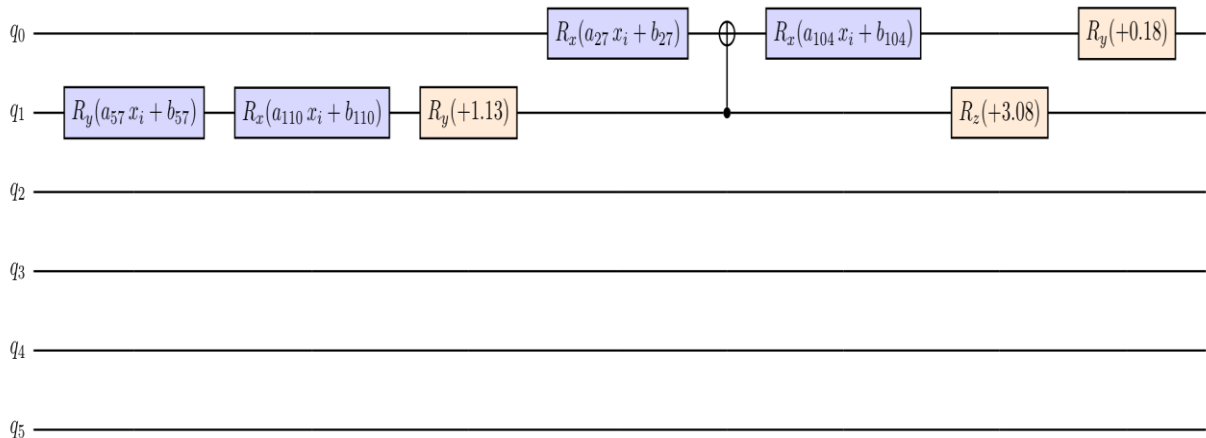
wine N=4 test_acc=1.000 encoder=reupload_euler



Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

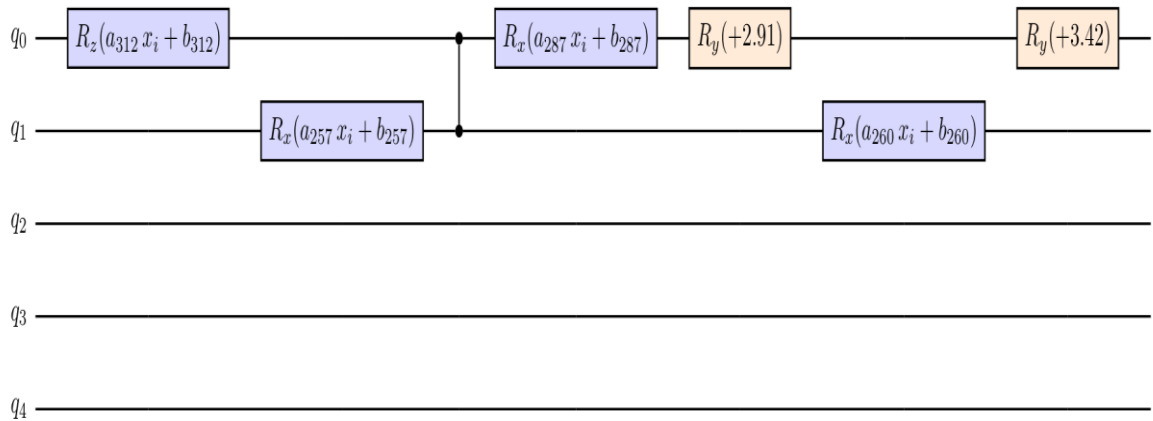
breast_cancer N=5 test_acc=0.974 encoder=linear_proj

Stage E v1: enc_ry/rx/rz (blue) interleaved with ansatz rotations (orange)



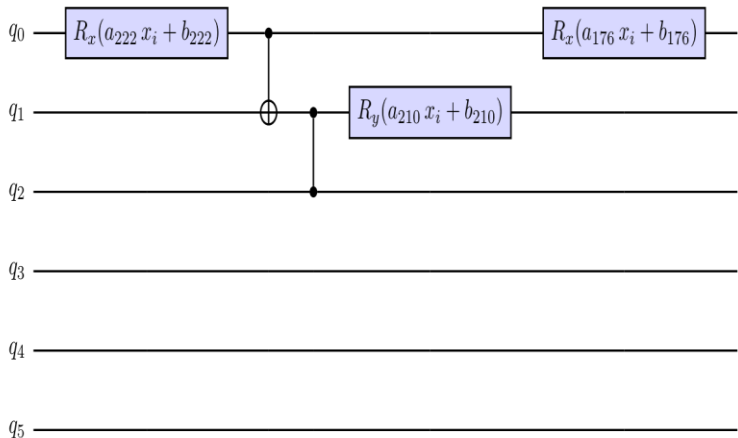
Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

iris N=6 test_acc=0.700



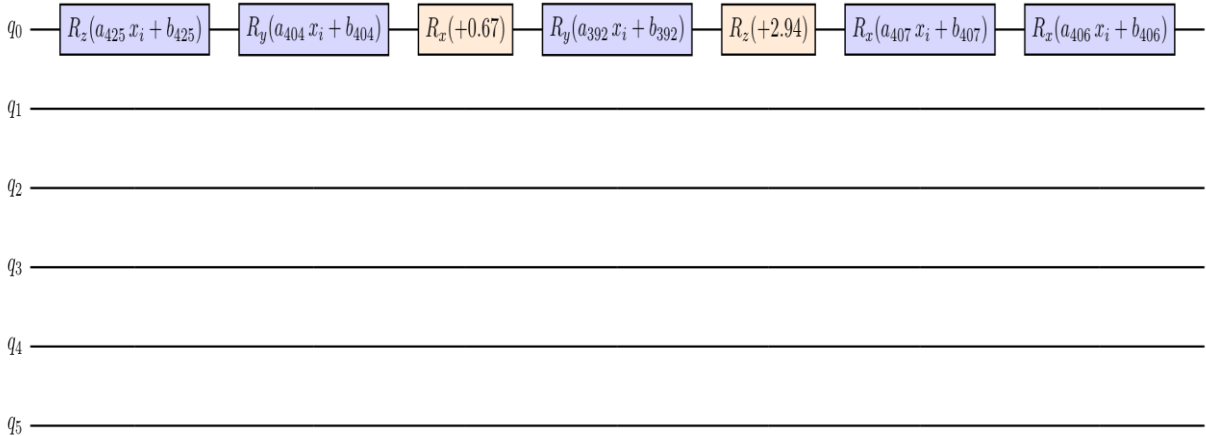
Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

seeds N=5 test_acc=0.833



Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

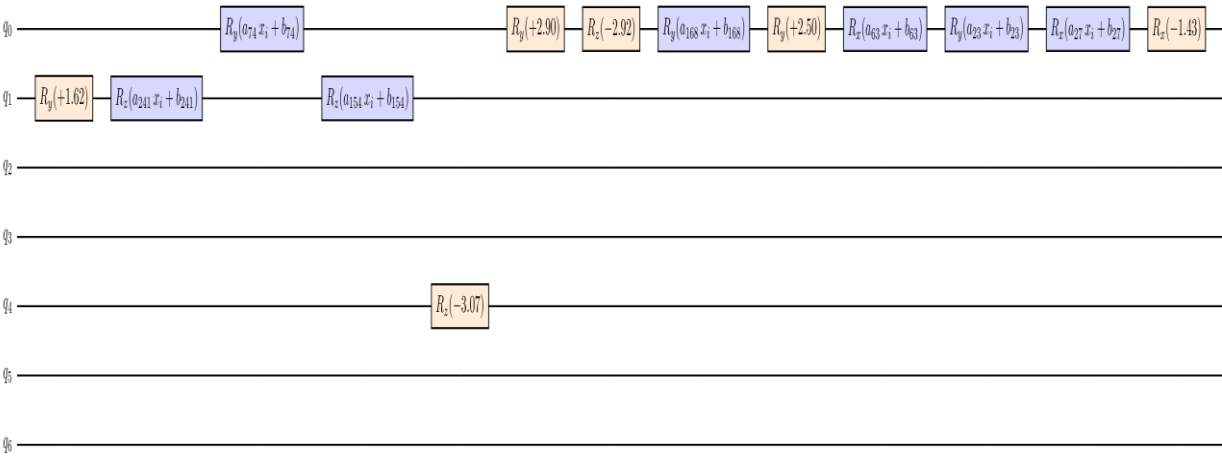
wine N=6 test_acc=0.778



Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

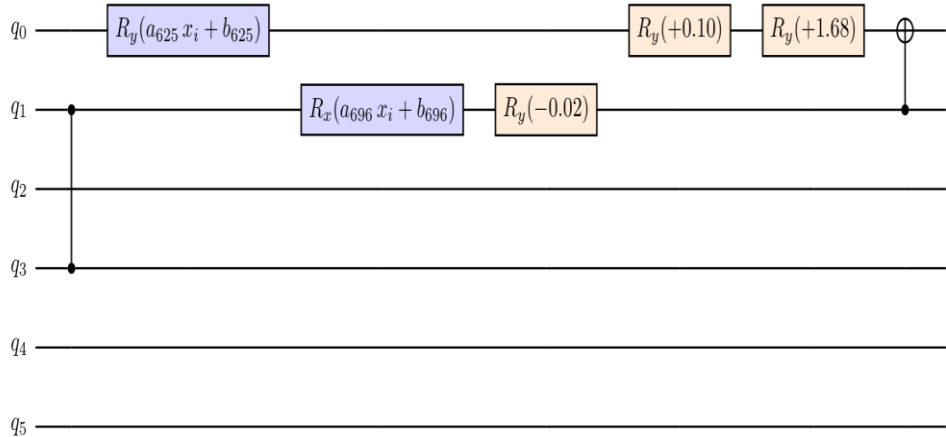
breast_cancer N=6 test_acc=0.851

Stage E v2: same as v1 with n_genomes=300



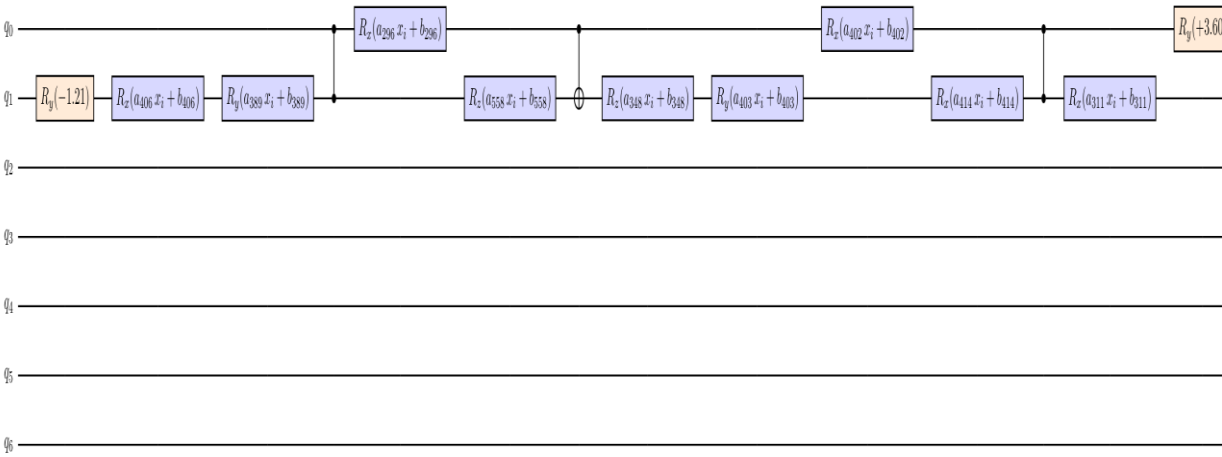
Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

iris N=7 test_acc=0.633



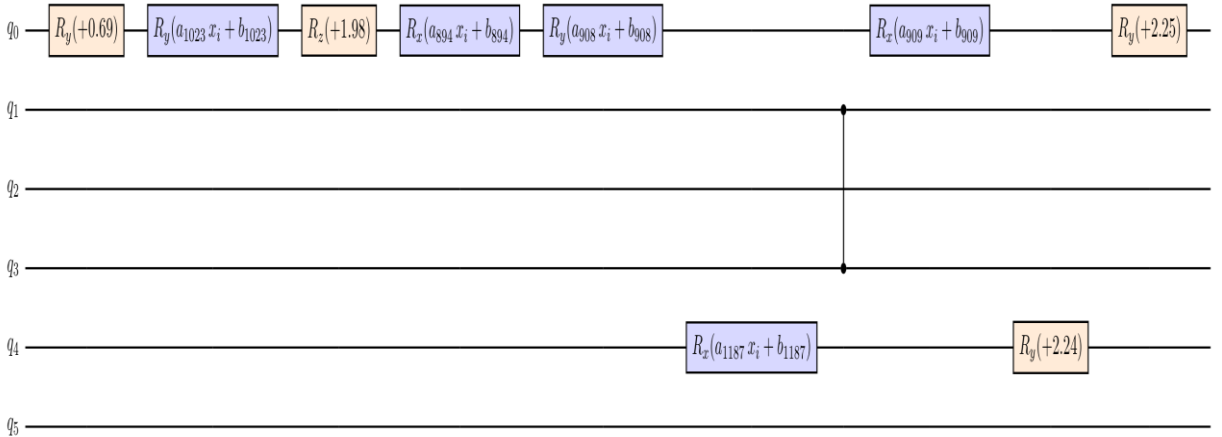
Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

seeds N=6 test_acc=0.833



Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

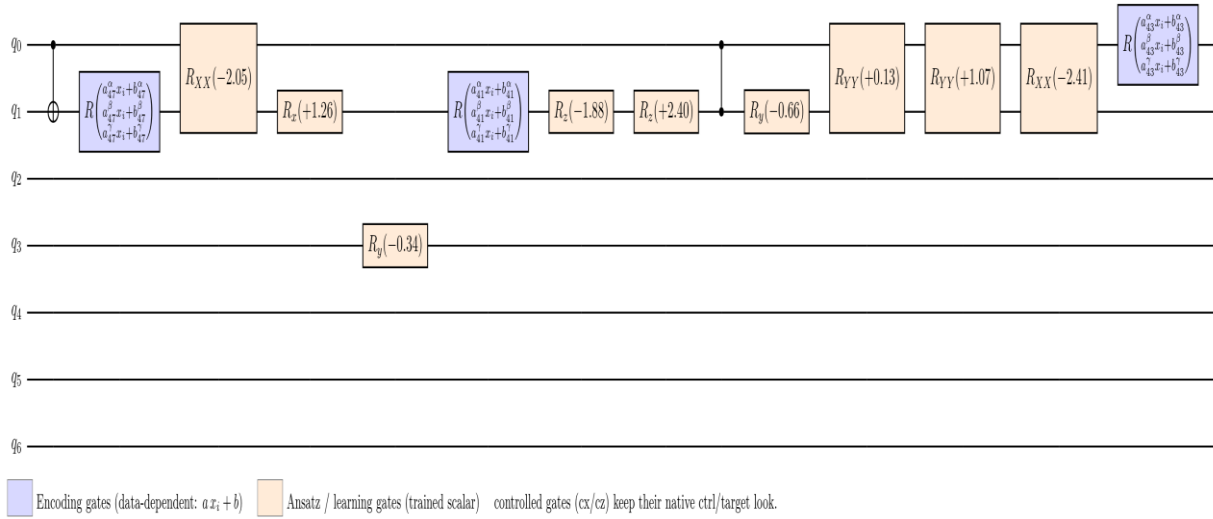
wine N=7 test_acc=0.806



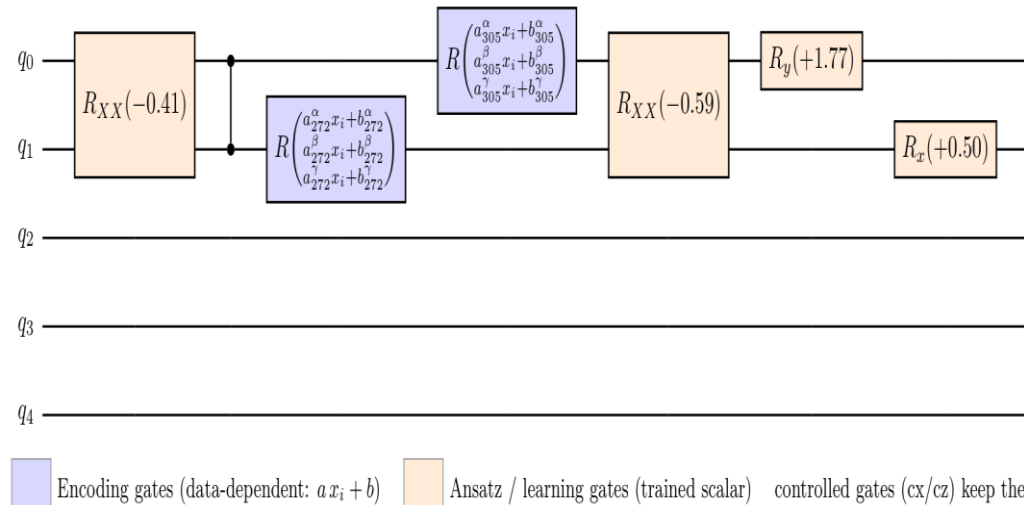
Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

breast_cancer N=6 test_acc=0.877

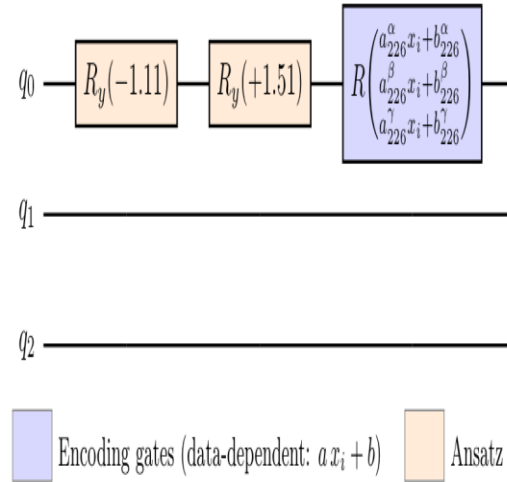
Stage F v1: enc_rot (blue) + scalar Ising entanglers (orange) -- interleaved



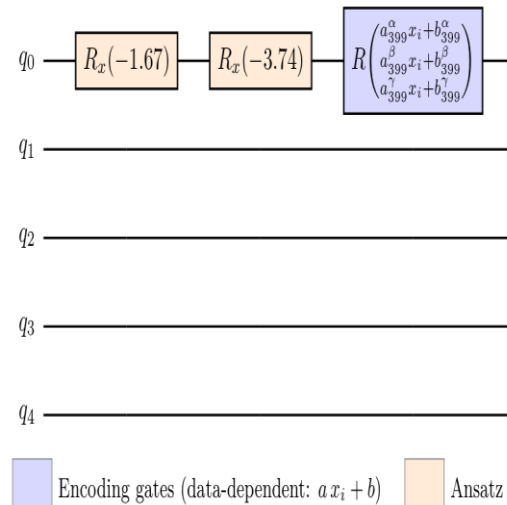
iris N=7 test_acc=0.733



seeds N=5 test_acc=0.833

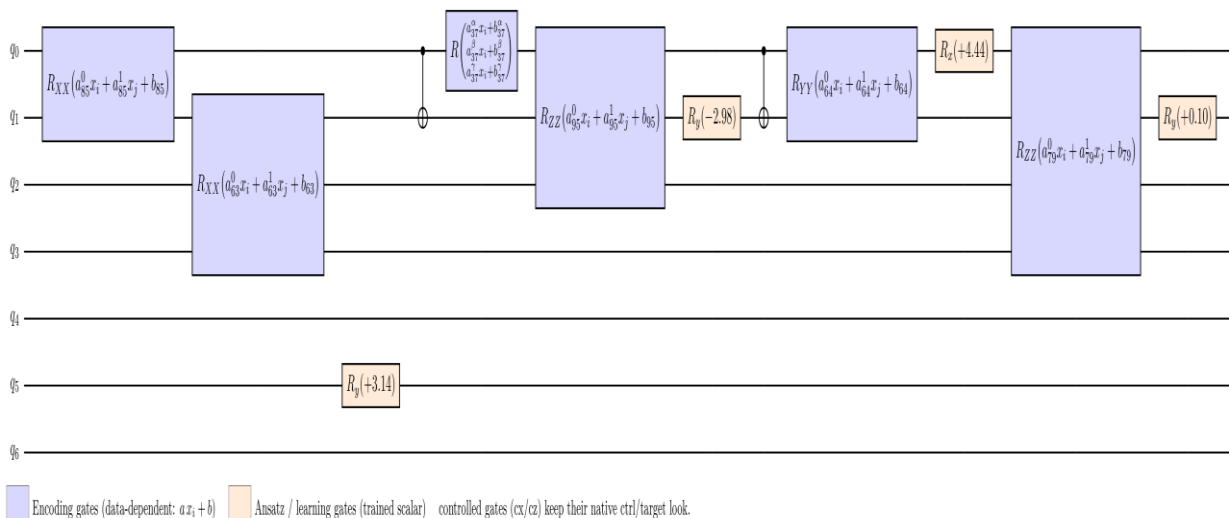


wine N=3 test_acc=0.806

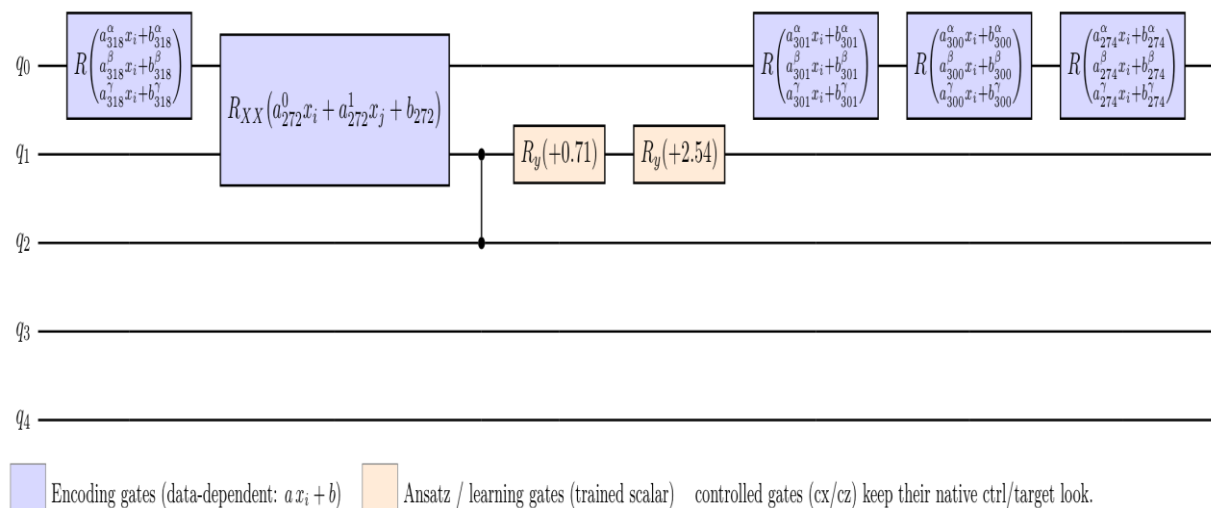


breast_cancer N=5 test_acc=0.842

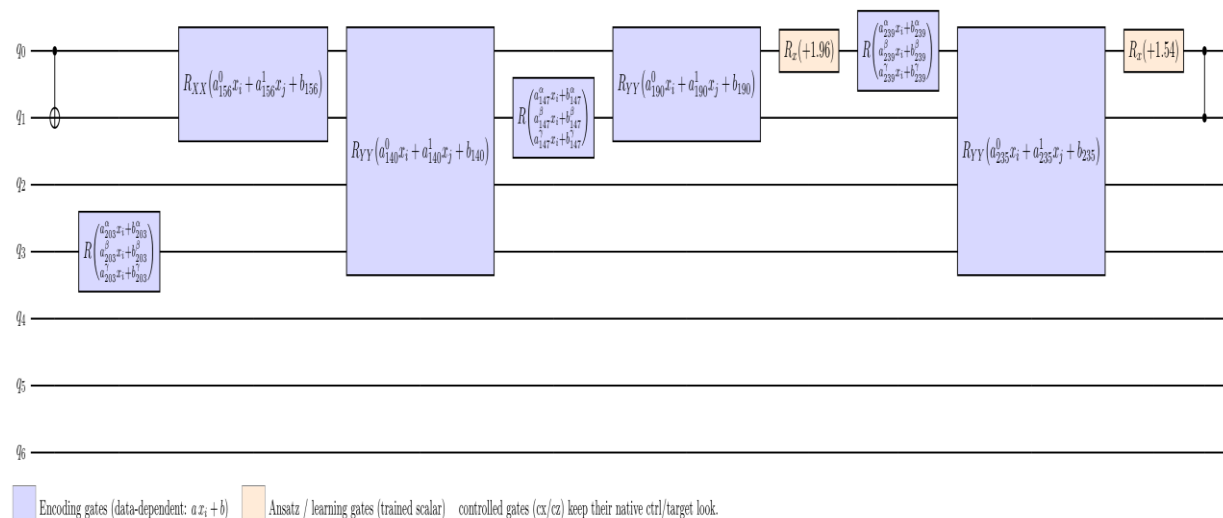
Stage F v2: enc_rot + enc_xx/yy/zz (blue) + ansatz rotations (orange) -- interleaved



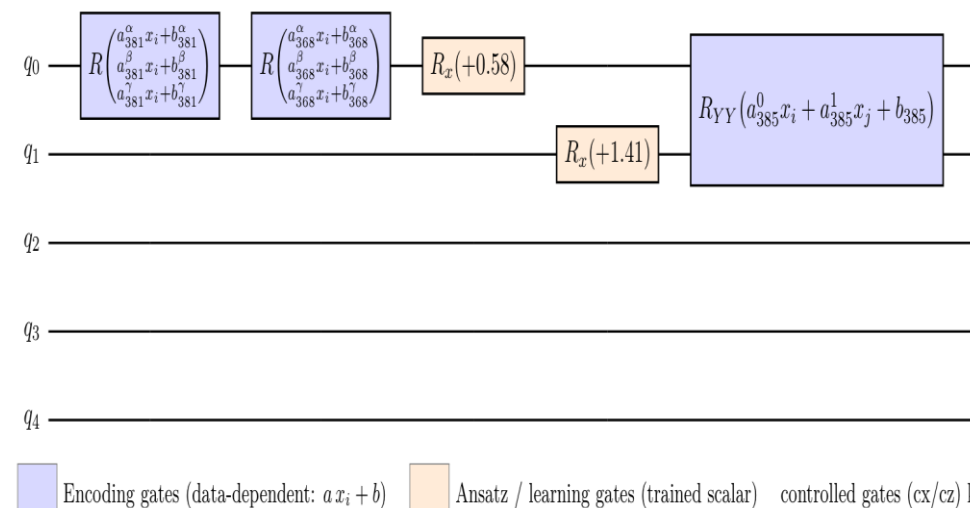
iris N=7 test_acc=0.967



seeds N=5 test_acc=0.857

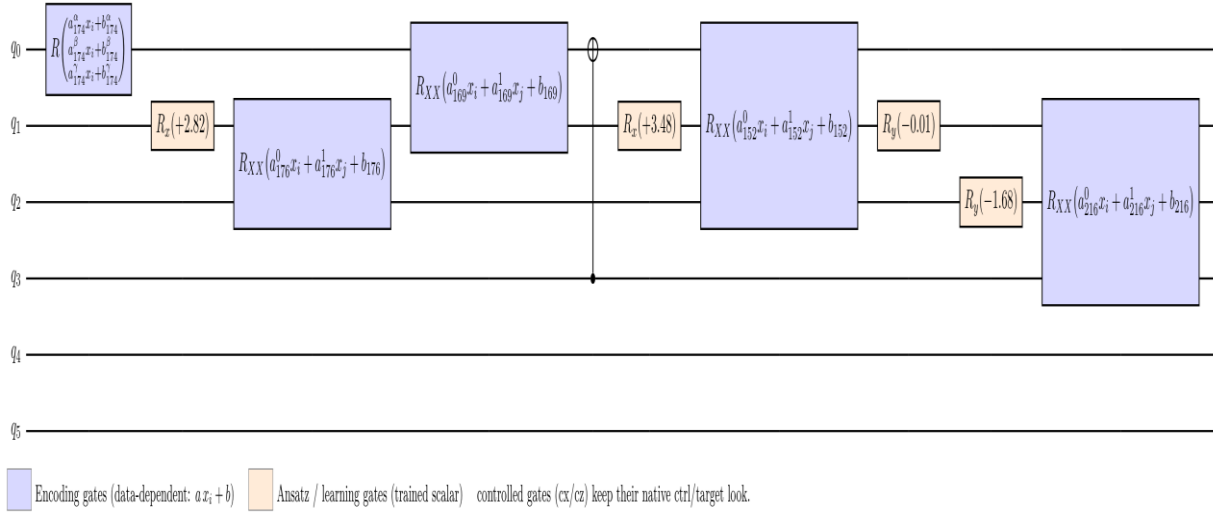


wine N=7 test_acc=0.917

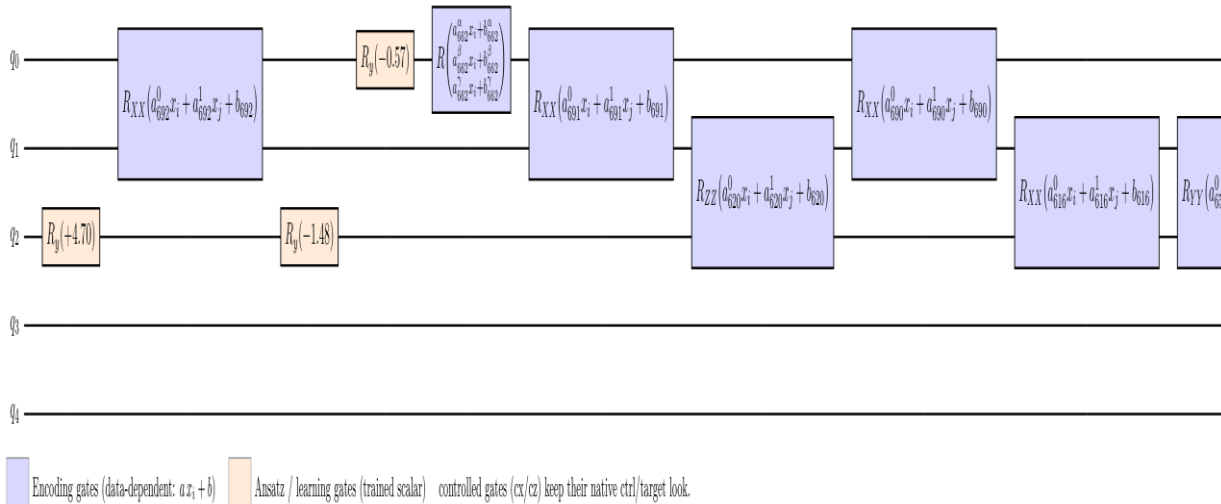


breast_cancer N=5 test_acc=0.877

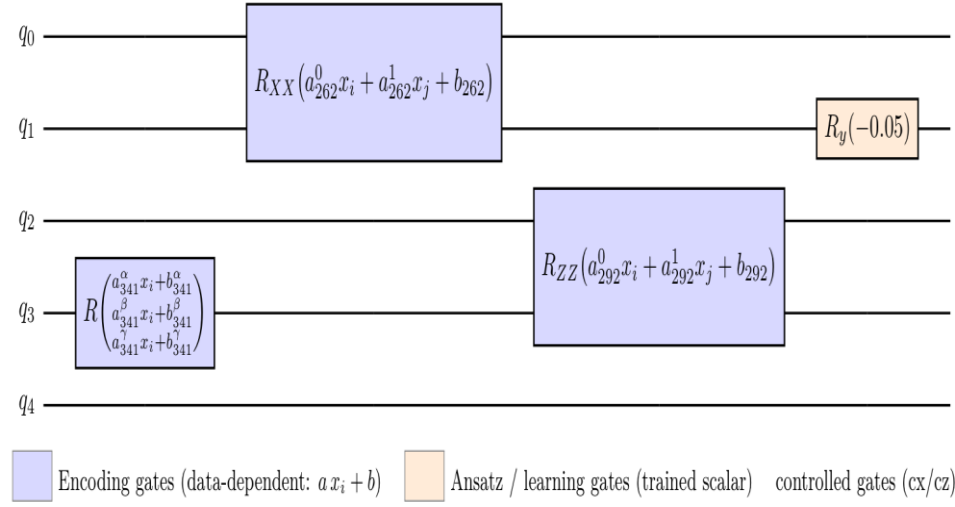
Stage F v2 3-seed champion



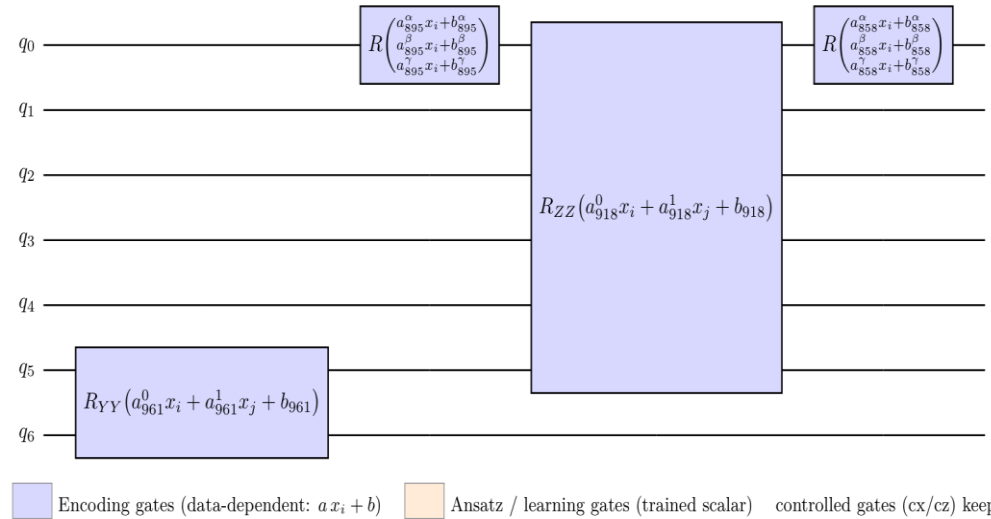
iris N=6 test_acc=1.000



seeds N=5 test_acc=0.833

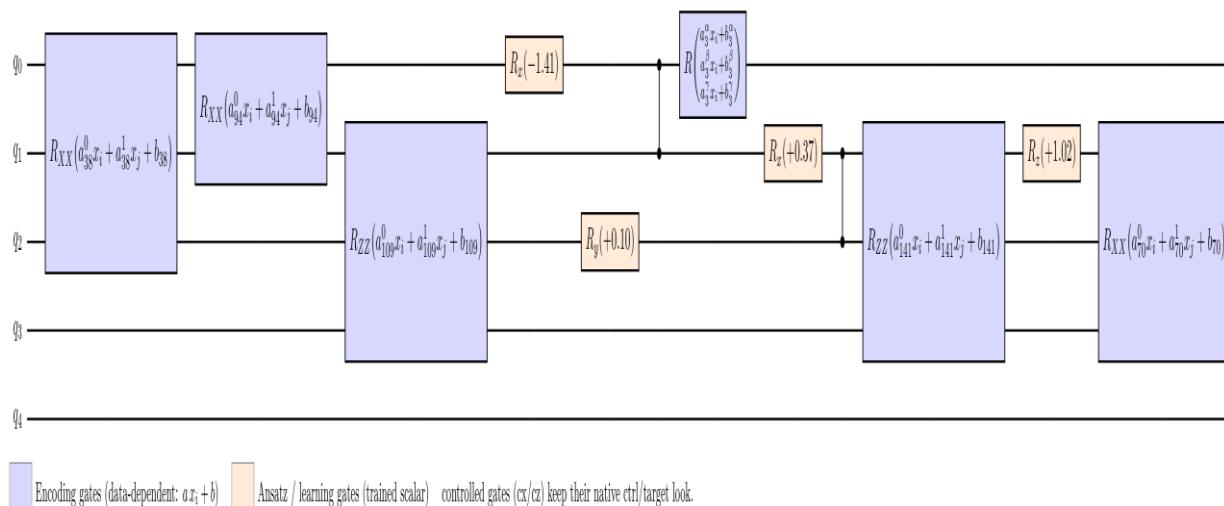


wine N=5 test_acc=0.833

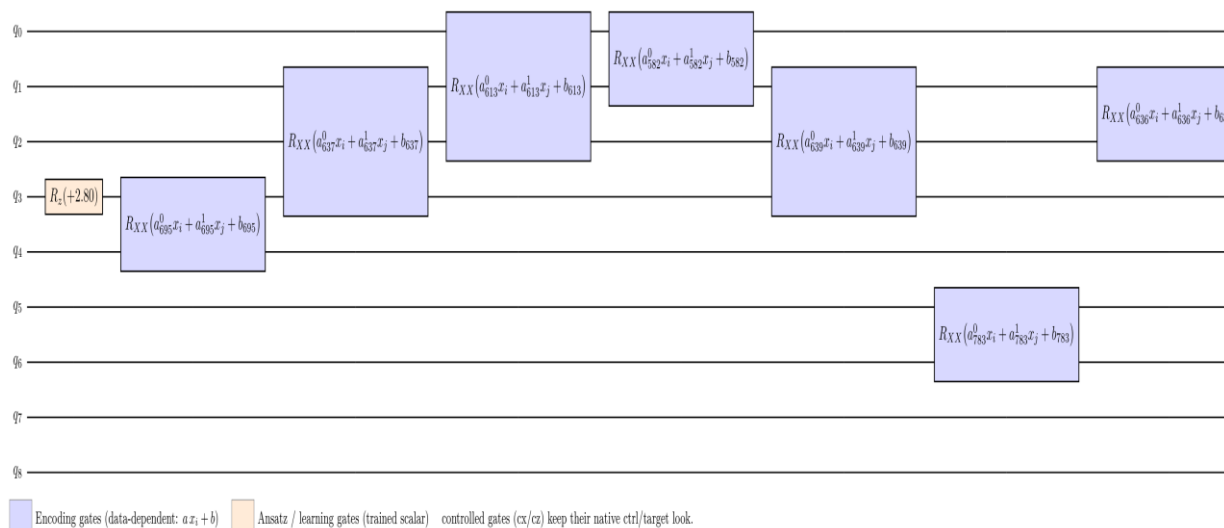


breast_cancer N=7 test_acc=0.886

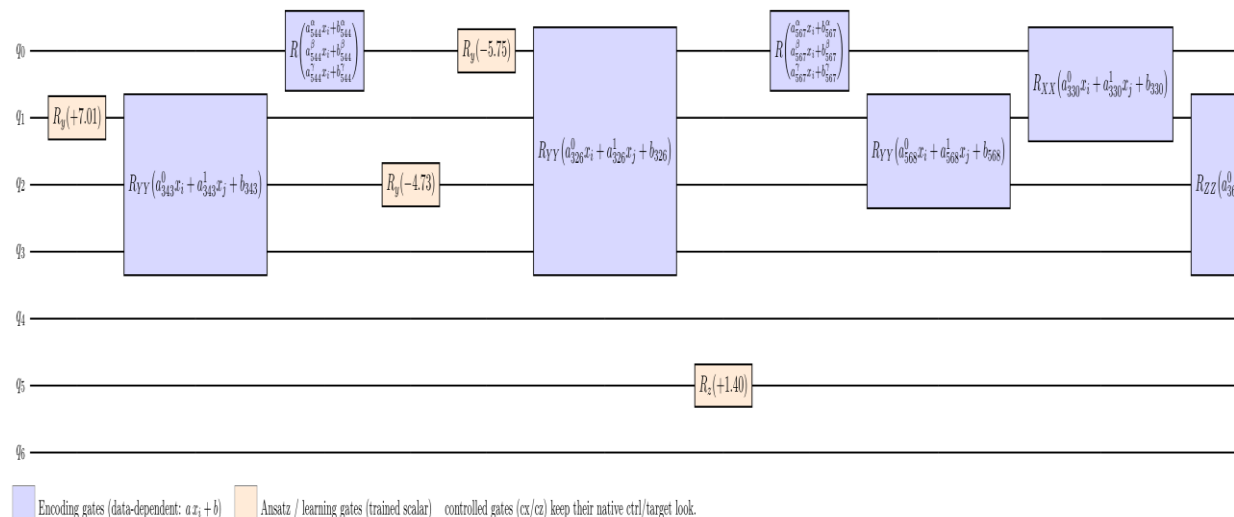
Stage F v2 budget bump (n_genomes=300)



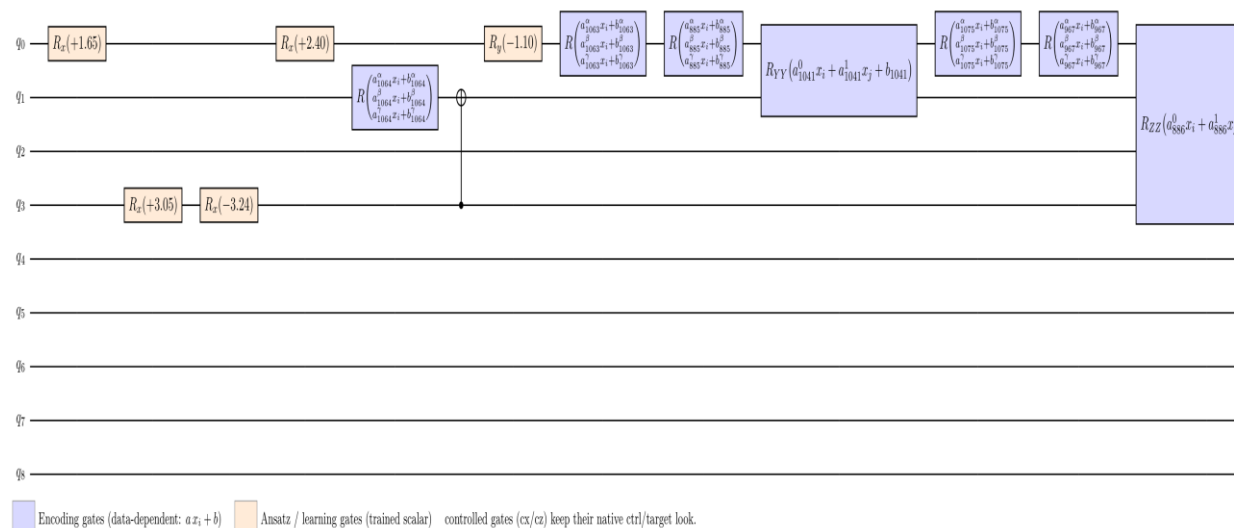
iris N=5 test acc=0.967



seeds N=9 test acc=0.857

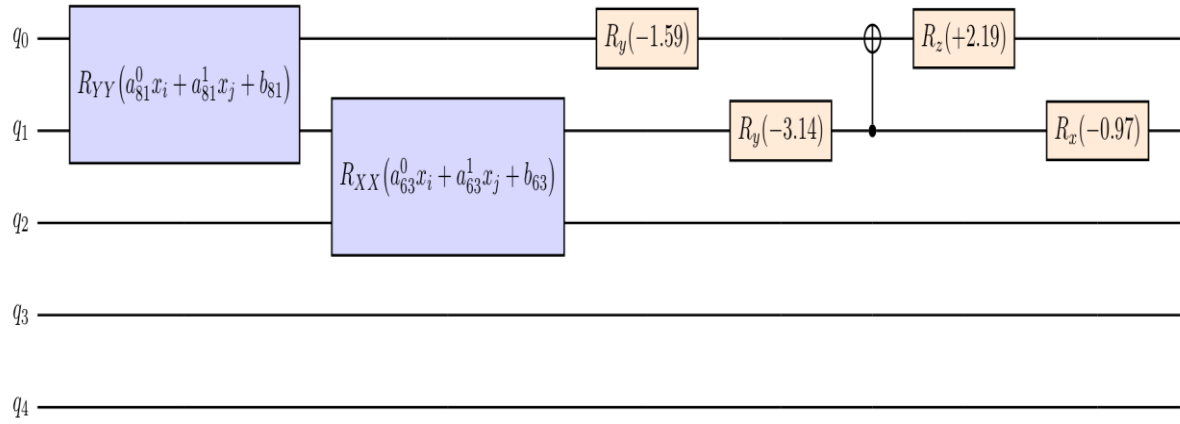


wine N=7 test acc=0.861



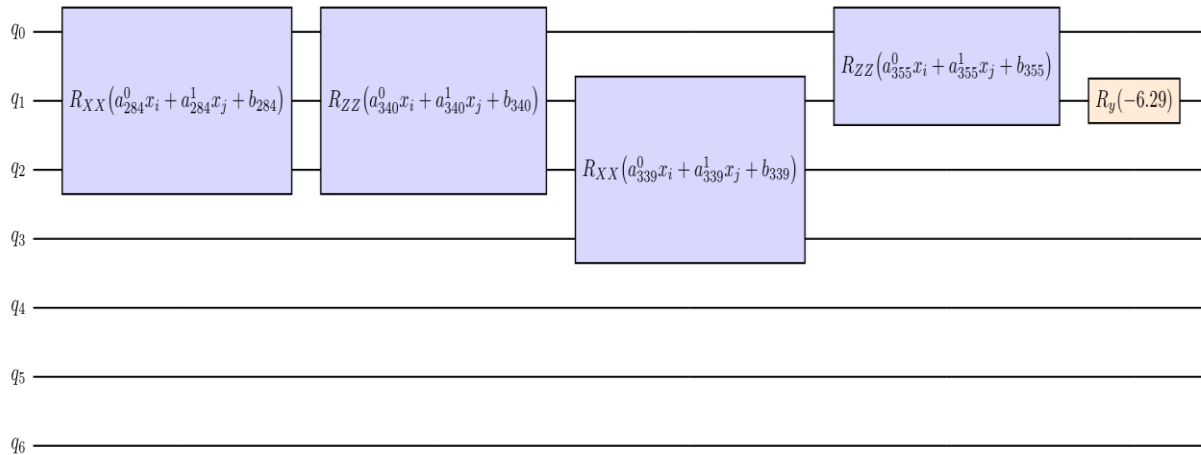
breast cancer N=9 test acc=0.912

Stage G: encoder block (blue) THEN ansatz block (orange) -- enforced by qnode



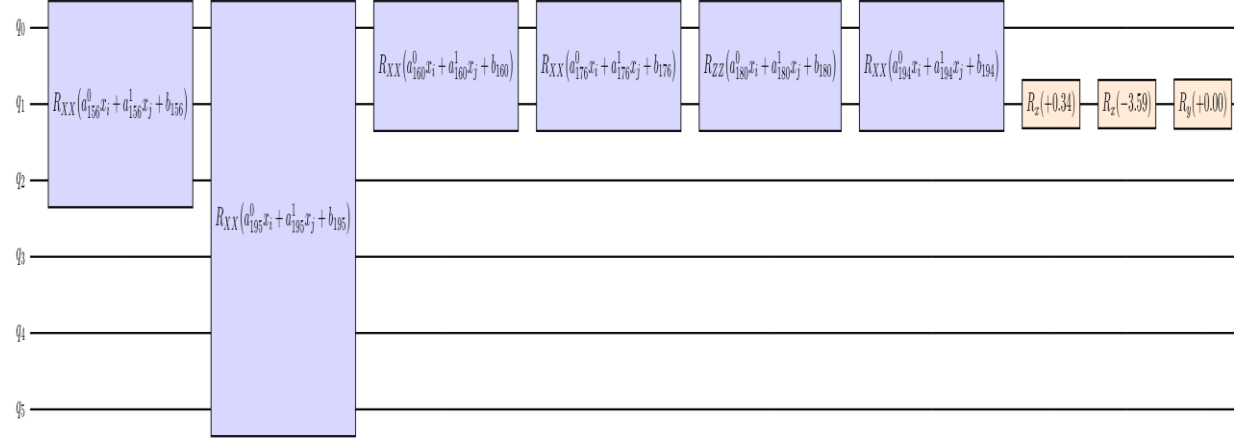
Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

iris N=5 test_acc=0.900



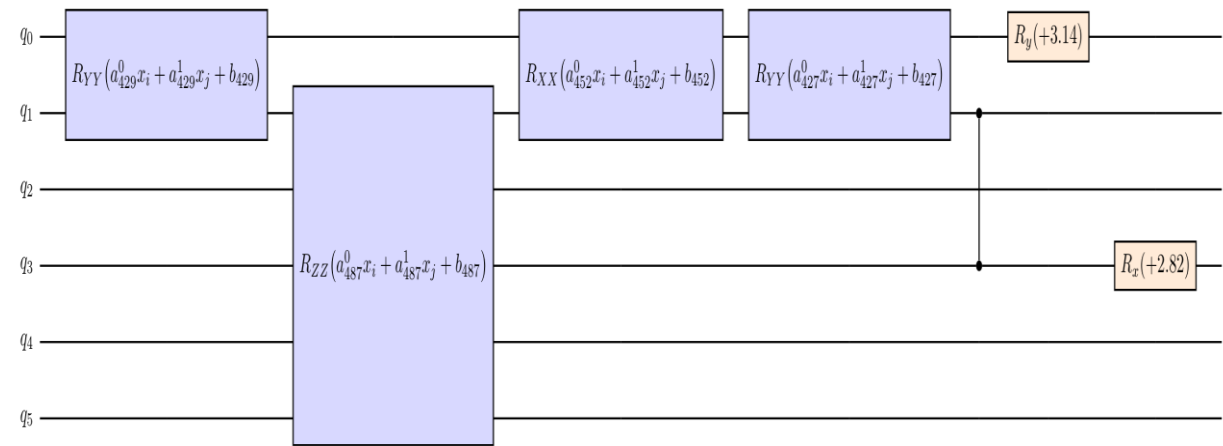
Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

seeds N=7 test_acc=0.810



Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

wine N=6 test_acc=0.861



Encoding gates (data-dependent: $a x_i + b$) Ansatz / learning gates (trained scalar) controlled gates (cx/cz) keep their native ctrl/target look.

breast_cancer N=6 test_acc=0.921